

Francisco Rau

Data & BI Analyst

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Summary

Data & BI analyst with an MSc in Data Science & Society and a BSc in Economics (Tilburg University). At Netvice I grew from data intern to Data & Process Analyst, building my own BI suite end to end — the centerpiece a forecasting model that turns planned mailing volume into the exact agent headcount needed on any future day, 500+ days ahead. I design the metric, not just the chart. Everything below is live and clickable at franciscorau.com.

Skills

Languages & tools: TypeScript, React, React Native (Expo), Node/Express, PostgreSQL, REST APIs, Git, Vitest, AI/agentic dev tooling, Excel

Methods: Statistical analysis, econometrics, forecasting, ETL & API integration, data visualization, machine learning

Domains: Business intelligence, operational analytics, KPI reporting, capacity planning

AI-assisted delivery: Agentic development with Claude Code and Replit

Experience

Data & Process Analyst — Netvice

May 2025 – Present · Tilburg, Netherlands

- Grew from data intern to Data & Process Analyst, working with increasingly senior stakeholders — ultimately on strategic projects directly with the company's founder.
- Established operational BI as a new, self-owned workstream — a first in the company's ten-year history — with full ownership from roadmap to daily use on the call-center floor.
- Fully self-directed: each of the systems below began as my own proposal — need identified, pitched, built, and shipped at pace.
- Forecasting model (the centerpiece): turns planned mailing volume into the exact number of agents needed on any future day — a cohort-based simulation of call attempts, pickup decay, and callback flows. Because it runs off the mailing calendar, agent demand is known 500+ days ahead, far beyond the notice period scheduling requires. Includes capacity-choke detection and a batch-splitting optimizer.
- Performance analytics: a percentile-based hourly pacing metric that makes agents comparable across campaigns with very different call lengths and success rates, combined with outcomes into a single performance index powering real-time TV rankings on the floor.
- Real-time operations layer: live dashboards wired to telephony webhooks (agent line state, idle detection, coverage views), plus a Daily Agent Planner, agent day reports, and a finance board (pilot).
- Introduced agentic AI development (Replit, Claude Code) as a delivery method — an early bet that let one analyst design, build, and ship production tooling end to end.
- Built everything configurable rather than hard-coded, so the tooling keeps working as campaigns and staff change.

Student Researcher — Philips

Sep 2022 – Dec 2022 · Eindhoven, Netherlands

- Researched OneBlade market penetration among students; built linear models estimating adoption of a blade-subscription service across student segments.

Data Entry — Von der Heide

Jan 2019 – Aug 2019 · Mexico City, Mexico

- Cleaned and migrated the company database to a new CRM system, safeguarding data integrity throughout the transfer.

Selected Projects

Market-Gap Analysis: Online-Booking Adoption in Mexico City

- End-to-end data pipeline measuring online-booking adoption across 10,605 beauty and wellness businesses in Mexico City. Includes Places API, headless browser, Streamlit

Mexico City: 722 structured API queries, deduplication, two independent booking detectors, and public-registry enrichment. Measured 6.9% adoption and delivered a 9,878-row qualified dataset (89% with direct contact details) as an Excel workbook and a Streamlit dashboard.

- Worked around a nationwide outage of the INEGI DENUA registry API by switching enrichment to its 43 MB bulk export, matched locally on industry code and location.

Weather Prediction-Market Research

2026 · Python, probability calibration, market microstructure

- Research system testing whether prediction markets price official weather observations efficiently: implied-probability extraction from order books, no-arbitrage coherence checks across temperature strike ladders, and measurement of repricing latency after each observation. Documented a null result — apparent edges did not survive fees and spread. Emphasis on rigorous testing and honest negative results.

Citania — Booking & Payments Platform

2025–2026 · TypeScript, PostgreSQL, React Native

- Solo-built multi-tenant booking and payments SaaS for beauty and wellness businesses in Mexico: 100+ integration tests, database-level slot exclusivity, and two-way Google Calendar sync.

Alpha Scout — Football Scouting & Valuation (beta)

2024–2026 · Python, statsmodels, Cloudflare

- Solo-built platform covering 23,445 players across 155 leagues: automated match-by-match data harvesting, a fair-value model ($R^2 = 0.74$ on live data) with a public per-league backtest, and calibrated 12-month probabilities. Forward-tested: players it flagged as undervalued were signed 1.7× more often than peers in the following transfer window.
- Built for honesty as a feature: the backtest is published per league including where the model is weak, and it declines to price players above €60M — outside its validated range.

Economics Paper on Erroneous Convictions in Mexico

2023 · Statistical analysis, survey data

- Worked with Roberto Hernández, co-creator of Mexico's national ENPOL survey, extracting statistical insights on false confessions and investigative practices from the raw survey data.

Education

MSc Data Science & Society (Business Track) — Tilburg University

2023 – 2024

BSc Economics — Tilburg University

2020 – 2023

International Baccalaureate — Greengates School, Mexico City

2010 – 2020

Languages

English (native), Spanish (native), Dutch (A2, actively learning)